Sri Chaitanya Educational Institutions | Infinity Learn

Sri Chaitanya Techno School
The right mentor for IIT (JEE), NEET, Olympiad & all Other Competitive exams

41 Years of Excellence 1986-2026

PERFECT SCORE.

Excellence has a permanent address
- Sri Chaitanya.

JEE Main 2026 (Session 1)



300
300

PASALA MOHITH
Appl.No: 260310432915
CLASSROOM STUDENT

Congratulations To Parents, Teachers & Staff

#TransformingYourDreamsIntoReality

www.srichaitanya.net 040-66 06 06 06

For More Join: @IITJEE_Advanced

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 25-03-2026**

Time: 09.00Am to 12.00Pm **GTM-28** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

Sri Chaitanya Educational Institutions

Infinity Learn



For More Join: @IITJEE_Advanced

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Rankers Academy JEE

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. If the system equations $x+y+z=5$, $x+2y+3z=9$, $x+3y+\alpha z=\beta$ has infinity many solutions then $\alpha^2+\beta^2$

1) 150 **2)** 154 **3)** 194 **4)** 154

2. If $\lim\limits_{x\to 0}\dfrac{axe^x-b\log(1+x)+cxe^{-x}}{x^2\sin x}=2$, then $a+b+c$ equal to

1) 18 **2)** 24 **3)** 30 **4)** 36

3. The value $\dfrac{{}^{100}C_{50}}{51}+\dfrac{{}^{100}C_{51}}{52}+.....+\dfrac{{}^{100}C_{100}}{101}$ is K. Then the value of $101K$ is.

1) 2^{101} **2)** $\dfrac{2^{101}}{101}$ **3)** 2^{100} **4)** $\dfrac{2^{100}}{100}$

4. The shortest distance between the lines $\bar{r}_1=\left(4\bar{i}-\bar{j}\right)+\lambda\left(\bar{i}+2\bar{j}-3\bar{k}\right)$ and $\bar{r}_2=\left(\bar{i}-\bar{j}+2\bar{k}\right)+\mu\left(2\bar{i}+4\bar{j}-5\bar{k}\right)$ is

1) $\dfrac{6}{\sqrt{5}}$ **2)** $\dfrac{\sqrt{6}}{5}$ **3)** $\dfrac{2}{\sqrt{3}}$ **4)** $\sqrt{\dfrac{5}{7}}$

5. Let the length of focal chord PQ of the parabola $y^2=12x$ be 15 units of the distance of PQ from the origin is d then $100\,d^2$ is equal to

1) 521 **2)** 740 **3)** 144 **4)** 720

6. The number of solution of the equations $2x+3\tan x=\pi$, $x\in[-2\pi,2\pi]-\left\{\neq\dfrac{\pi}{2},\pm\dfrac{3\pi}{2}\right\}$ is

1) 6 **2)** 5 **3)** 4 **4)** 3

Sri Chaitanya Educational Institutions Infinity Learn 41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 JEE Advanced 2023 RANK 1 NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

7. If z is a complex number such that $|z| \geq 3$, then the minimum value of $\left|z + \frac{(6+8i)}{2z}\right|$ is

1) 2 **2)** $\frac{4}{3}$ **3)** $\frac{5}{3}$ **4)** 10

8. If α and β are the roots of the equation $x^2 - x + 2 = 0$ and $S_n = 2024\alpha^n + 2025\beta^n$, then

1) $S_{10} = 2S_9 - S_8$ **2)** $S_{10} = S_9 - 2S_8$ **3)** $2S_{10} = S_9 - S_8$ **4)** $S_{10} = S_9 + 2S_8$

9. Let the Range of the function $f(x) = \frac{1}{\sin x + \cos x - 2}$, $x \in R$, be $[a,b]$, if α and β are the A.M and G.M. of a and b respectively, then $\alpha^2 - \beta^2$ is equal to

1) $\frac{1}{\sqrt{2}}$ **2)** 1 **3)** $\frac{1}{2}$ **4)** -2

10. If $\bar{e} = l\bar{i} + m\bar{j} + n\bar{k}$ is a unit vector

Statement – I: The maximum value of $lm + mn + nl$ is 1

Statement – II: If $l = \frac{1}{3}$ then maximum value of lmn is $\frac{4}{3}$

1) Statement–I is true, statement–II is false

2) Statement–I is false, statement–II is true

3) Both statement–I and statement–II are true

4) Both statement–I and statement–II are false

11. Let $A = \{(\alpha,\beta) \in R \times R : |(\alpha - 1)| \leq 4 \text{ and } |\beta - 5| \leq 6\}$ and

$B = \{(\alpha,\beta) \in R \times R : 16(\alpha - 2)^2 + 9(\beta - 6)^2 \leq 144\}$

1) $B \subset A$ **2)** $A \cup B = \{(x,y) : -4 \leq x \leq 4, -1 \leq y \leq 11\}$

3) Neither $A \subset B$ nor $B \subset A$ **4)** $A \subset B$

12. Let A matrix order 3×3 such that $|A| = 1$ let $B = 2A^{-1}$ and $C = \frac{adjA}{\sqrt[3]{2}}$ then the value of

$|AB^2C^3| =$ (where $|A|$ represent $\det A$)

1) 6 **2)** 8 **3)** 10 **4)** 12

Sri Chaitanya Educational Institutions Infinity Learn 41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 JEE Advanced 2023 RANK 1 NEET 2023 RANK 1
Rankers Academy JEE
For More Join: @IITJEE_Advanced

13. Let $\bar{a}=2\lambda\hat{i}+2\hat{j}-2\hat{k}$ and $\bar{b}=6\hat{i}-2\hat{j}+4\hat{k}$ be two vectors and λ be an integer. Let $'\bar{c}'$ be a non zero vector such that $'\bar{c}'$ is parallel to $\left(\bar{a}+\bar{b}\right)$, $\left(\bar{a}.\bar{c}\right)=-68$ and $\bar{b}\cdot\bar{c}=-80$, then the value of $\left[\dfrac{\left|\bar{c}\times\bar{b}\right|^2}{\left|\bar{a}\right|^2}\right]$ is ____

Where $[\cdot]$ is the greatest integer function.

1) 10 **2)** 11 **3)** 118 **4)** 120

14. Let $y=x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x\geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point $(2,3)$ and is farthest from the center of C_2, is $x+ay+b=0$, then $a+b$ is equal to:

1) 4 **2)** 3 **3)** 2 **4)** 0

15. The natural number $'c'$ for which $\sum_{r=1}^{n} f(c+r)=16\left(2^n-1\right)$, where the function satisfies the relation $f(x+y)=f(x)\cdot f(y)$, for all $x,y\in N$, and $f(1)=2$.

1) 1 **2)** 2 **3)** 3 **4)** 4

16. If the area of the region $\left\{(x,y):1+x^2\leq y\leq \min(x+7,11-3x)\right\}$ is A then $6A$ is equal to

1) 50 **2)** 100 **3)** 49 **4)** 46

17. The mean and standard deviation of 100 observations are 40 and 5.1 respectively, by mistake one observation is taken as 50 instead of 40. If the correct mean and the correct standard deviation are μ and σ respectively. Then $100(\mu+\sigma)$ is equal to

1) 4490 **2)** 3692 **3)** 4390 **4)** 4400

18. A bag contains 10 balls, whose colours are either red or green. 6 balls are drawn at random without replacement and it is found that 3 balls are red and other 3 balls are green. The probability that bag contains equal number of red and green balls is ____

1) $\frac{7}{2}$ **2)** $\frac{10}{37}$ **3)** $\frac{10}{33}$ **4)** $\frac{7}{33}$

Sri Chaitanya Educational Institutions Infinity Learn 41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 JEE Advanced 2023 RANK 1 NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

19. Let the straight line $y=2x$, touch a circle with centre $(0,\alpha)$, $\alpha>0$ and radius r at a point A_1, let B_1 be the point on the circle such that the line segments A_1,B_1 is diameter of the circle, let $\alpha+r=5+\sqrt{5}$. Match the following:

Column – I		Column – II	
P)	α equals	I)	$(-2,4)$
Q)	r equals	II)	$\sqrt{5}$
R)	A_1 equals	III)	$(-2,6)$
S)	B_1 equals	IV)	5
		V)	$(2,4)$

The correct option

1) $P-IV,Q-II,R-I,S-III$ **2)** $P-II,Q-IV,R-I,S-III$

3) $P-IV,Q-II,R-V,S-III$ **4)** $P-II,Q-IV,R-III,S-V$

20. Let f be a differentiable function satisfying $f(x)=1-2x+\int_0^x e^{(x-t)}f(t)\,dt,\ x\in R$ and let $g(x)=\int_0^x (f(t)+2)^{15}(t-4)^6(t+12)^{17}\,dt,\ x\in R$. If p and q are respectively the points of local minima and local maxima of g, then the value of $\frac{|p+q|}{3}$ is equal to _____

1) 3 **2)** 6 **3)** 9 **4)** 12

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

Sri Chaitanya Educational Institutions Infinity Learn 41



For More Join: @IITJEE_Advanced

21. The number of integral solution are there to $x+y+z+t=29$, when $x\geq 1,\ y>1,\ z\geq 3$ and $t\geq 0$ is

22. Let one focus of the hyperbola $H:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ be at $\left(\sqrt{10},0\right)$ and corresponding direction be $x=\frac{9}{\sqrt{10}}$. If e and l respectively are the eccentricity and length of latus rectum of H. Then $27\left(e^2+l\right)$ is equal to

23. Let the domain if the function $f(x)=\log_2\log_4\log_6\left(3+4x-x^2\right)$ be (a,b). If $\int_0^{b-a}\left[x^2\right]dx=p-\sqrt{q}-\sqrt{r},\ p,q,r\in N,\gcd(p,q,r)=1$, where [.] is G.I.F. then $\frac{p+r}{q}$ is equal to

24. If the solution of differential equation $(2x+3y-1)dx+(4x+6y-5)dy=0,\ y(0)=2$ is $\alpha x+\beta y+9\ln(2x+3y-\gamma)=12$ then $3\alpha+2\beta+\gamma$ is

25. The value of $3\sum_{n=1}^{\infty}\left[\frac{1}{\pi}\sum_{k=1}^{\infty}\cot^{-1}\left(1+2\sqrt{\sum_{r=1}^{k}r^3}\right)\right]^n$ is

Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. Match List – I with List – II

List – I		List – II	
A)	Planck's constant (h)	I)	$M^1L^2T^{-2}$
B)	Momentum (P)	II)	M^1L^2
C)	Moment of Inertia (I)	III)	$M^1L^2T^{-1}$
D)	Torque (τ)	IV)	$M^1L^1T^{-1}$

Choose the correct answer from the options given below.

1) $A-III, B-I, C-IV, D-II$ **2)** $A-III, B-IV, C-II, D-I$

3) $A-IV, B-III, C-II, D-I$ **4)** $A-II, B-III, C-IV, D-I$

27. A cubical block of density $\rho_b = 800 kgm^{-3}$ floats in a liquid of density $\rho_l = 1200 kgm^{-3}$. If the height of block is $H = 10 cm$, then the height of the submerged part $(in\ cm)$ is ____

1) 7.3 **2)** 5.3 **3)** 6.3 **4)** 9.3

28. A bulb is placed at a depth of $2\sqrt{7}\,m$ in water and a floating opaque disc is placed over the bulb so that the bulb is not visible from the surface. The minimum radius of the disc is (refractive index of water $\frac{4}{3}$)

1) 6 m **2)** 12 m **3)** 8 m **4)** 4 m

29. In an experiment the values of two spring constants were measured as $K_1 = (15 \pm 0.2) N/m$ and $K_2 = (10 \pm 0.3) N/m$. If these springs are connected in parallel, then the percentage error in the equivalent spring constant is

1) 1% **2)** 2% **3)** 2.5% **4)** 3%

Sri Chaitanya Educational Institutions Infinity Learn


For More Join: @IITJEE_Advanced

30. When a disc rotates with uniform angular velocity, which of the following is true?

A) The sense of rotation remain same

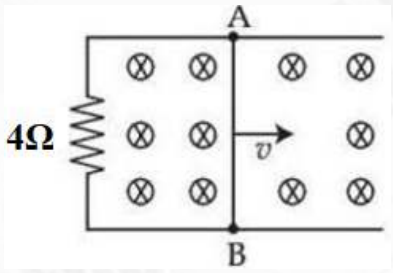
B) The orientation of the axis of rotation remains same

C) The speed of rotation changed

D) The angular-acceleration is non-zero and remains same

1) A is correct **2)** A & B correct **3)** A & C correct **4)** C & D correct

31. A $1m$ long metal rod AB completes, the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10T. If the resistance of the total circuit is 4Ω, then the force needed to move the rod towards right with constant speed (v) of $2m/s$ is ____N (Rod has no resistance)



1) 7.5×10^{-2} **2)** 5×10^{-2} **3)** 5×10^{-3} **4)** 7.5×10^{-3}

32. The horizontal range of a projectile fired at an angle 15° is $40\ m$. If it is fired with the same speed at angle of 45°, Its range will be

1) $60\ m$ **2)** $71\ m$ **3)** $100\ m$ **4)** $80\ m$

33. **Statement – I:** In a meter bridge experiment, null point for an unknown resistance is measured. Now, the unknown resistance is put inside an enclosure maintained at a higher temperature. The null point can be obtained at the same point as before by decreasing the value of the standard resistance

Statement – II: Resistance of a metal increases with the increases in temperature

1) Statement–I is true, statement–II is true, statement – II is a correct explanation for statement – I

2) Statement–I, statement–II is true, statement–II is not correct explanation for statement–I

3) Statement–I is true, Statement-II is false

4) Statement–I is false, Statement–II is true

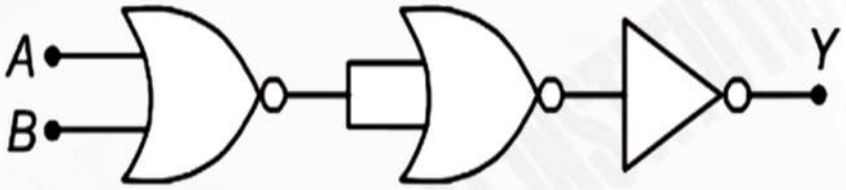
Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

34. A block of mass 10kg starts sliding on a rough horizontal surface with an initial velocity of $4.9\,m/s$. The coefficient of friction between the surface and block is 0.5.The distance covered by the block before coming to rest is: (Use $g = 9.8\,m/\sec^2$)

1) $4.9m$ **2)** $9.8m$ **3)** $2.45m$ **4)** $12.5m$

35. The given electrical network is equivalent to



1) NOT gate **2)** AND gate **3)** OR gate **4)** NOR gate

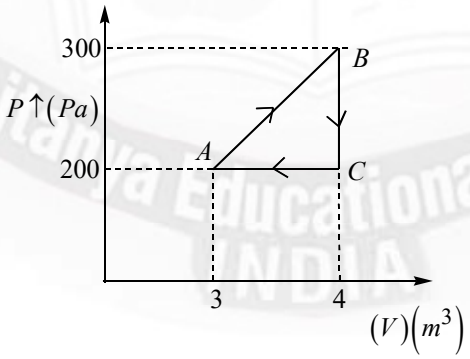
36. A body of mass m is taken from earth surface to the height h equal to thrice the radius of earth (R_e), the increase in potential energy will be

1) $3mgR_e$ **2)** $\frac{1}{3}mgR_e$ **3)** $\frac{3}{4}mgR_e$ **4)** $\frac{2}{3}mgR_e$

37. A series combination of R, L, C is connected to an a.c source. If the resistance is 3Ω and the reactance is $4\,\Omega$, what is the power factor of the circuit ?

1) 1 **2)** 0.6 **3)** 0.8 **4)** 0.5

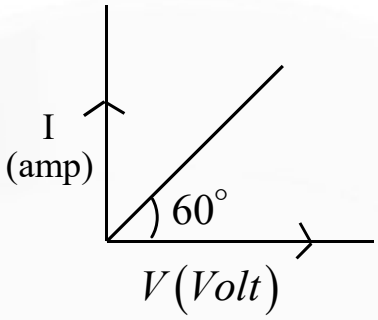
38. A thermodynamic system is taken through the cyclic process ABC as shown in the figure. The total workdone by the system during the cycle ABC is _____ J.



1) 50 **2)** 100 **3)** 250 **4)** 200

Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

39. Current flowing through a circuit changes with voltage as shown in the figure. Then the resistance of the circuit is



1) $\sqrt{3}\,\Omega$ **2)** $\frac{1}{\sqrt{3}}\,\Omega$ **3)** $1\,\Omega$ **4)** $\frac{\sqrt{3}}{2}\,\Omega$

40. **Assertion (A):** A simple pendulum is taken to a planet of mass and radius, 2 times and 4 times respectively, than the earth. The time period of the pendulum remain same on earth and the planet

Reason (R): The mass of the pendulum remains unchanged at earth and the other planet. In the light of the above statements, choose the correct answer from the options given below.

1) Both (A) and (R) are true and (R) is the correct explanation of (A)

2) (A) is false but (R) is true

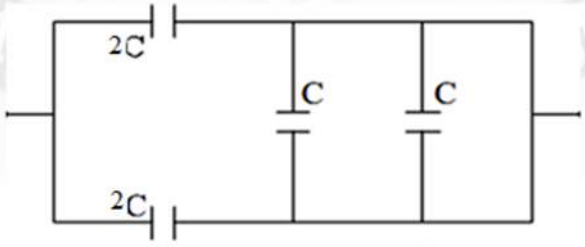
3) (A) is true but (R) is false

4) Both (A) and (R) are true and (R) is NOT the correct explanation of (A)

41. An electron is released with a velocity of $5\times10^{6}\,ms^{-1}$ in an electric field of $10^{3}\,NC^{-1}$ which has been applied so as to oppose its motion. What distance would the electron travel before it is brought to rest?

1) $0.07cm$ **2)** $0.07m$ **3)** $0.14\ mm$ **4)** $0.014mm$

42. The equivalent capacitance of the combination shown is



1) $\frac{C}{2}$ **2)** $4C$ **3)** $2C$ **4)** $\frac{5}{3}C$

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

43. **Statement – I:** Though light of single frequency (mono chromatic light) is incident on a metal, the kinetic energies of emitted photo electrons are same

Statement – II: The energy of electrons just after they absorb photons incident on metal surface may be lost in collision with other atoms in the metal before the electron is ejected out of metal

1) Statement–I is true, statement–II is true, statement–II is a correct explanation for statement–I

2) Statement–I, statement–II is true, statement–II is not correct explanation for statement–I

3) Statement – I is true, Statement-II is false

4) Statement – I is false, Statement – II is true

44. The magnetic field at the centre of a circular current carrying loop of area A is B. The magnetic moment of the loop is M. Then

1) $M=\frac{BA}{\mu_0}\sqrt{A}$ **2)** $M=\frac{BA^2}{\mu_0\pi}$ **3)** $M=\frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$ **4)** $M=\frac{BA\sqrt{A}}{\mu_0\pi}$

45. On disintegration of one atom of U^{235} the amount of energy obtained is $200MeV$. The power obtained in a reactor is 10 megawatt. How many atoms are undergoing fission per second?

1) 3.125×10^{17} **2)** 3.125×10^{16} **3)** 200 **4)** 5×10^{3}

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

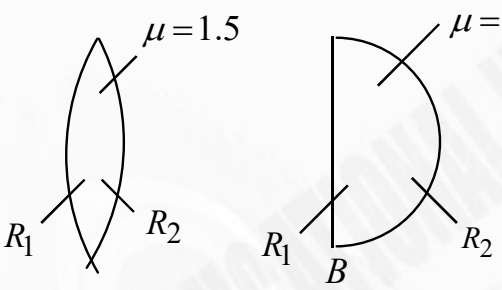
46. A seconds pendulum clock has a steel wire, the clock is calibrated at $20^{\circ}C$. How much time in seconds does the clock lose (or) gain in one week, when the temperature

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

increased to $40^{\circ}C$?(Round off to nearest integer) (Coefficient of linear expansion $\alpha = 1.2\times10^{-5}/^{\circ}C$)

47. Two lenses one biconvex and other plano convex have same magnitude of power. The refractive indices of their materials are 1.5 and 1.6 respectively. If the radii of curvature of the lenses are as shown, then the ratio $\frac{R_1}{R_2} = x:1$, where x is



48. The ratio of the deBroglie wavelengths of the electron in the 2nd excited state of hydrogen atom and the 1st excited state of Li^{2+} ion is $\frac{x}{2}$, where x is

49. Two identical solid spheres each of mass $4\,kg$ and radius $10\,cm$ are kept in contact, then the moment of inertia of this pair of spheres about the tangent passing through point of contact is $x\times10^{-3}\ kg\,m^2$, where x is_____

50. Young's double slit experiment is performed inside water $\left(\mu = \frac{4}{3}\right)$ with light of f requency $6\times10^{14}\,Hz$. If the slits are separated by $0.2\ mm$ and the screen kept $1m$ from the slits. Then the fringe width ____ (in mm) (Round off to nearest integer).

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

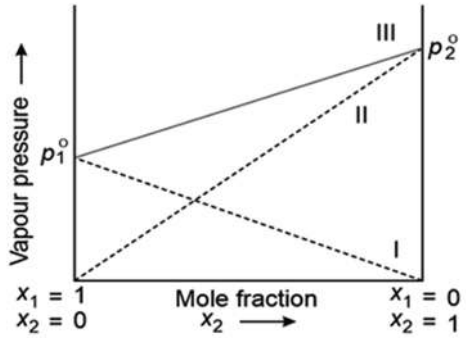
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Which of the following set of quantum number is not allowed?

1) $n=4,\ell=2,m\ell=0,s=+\frac{1}{2}$ **2)** $n=3,\ell=1,m\ell=-2,s=+\frac{1}{2}$

3) $n=3,\ell=3,m\ell=-3,s=-\frac{1}{2}$ **4)** $n=3,\ell=0,m\ell=0,s=-\frac{1}{2}$

52. The diagram given below represents the vapour pressure and mole fraction of an ideal solution of component 1 and 2



Which of the following statements is true about the diagram?

1) Component 1 is more volatile than component 2

2) Component 1 is less volatile than component 2

3) Component 1 is equally volatile than component 2

4) None of the above is true

53. Given below are two statements

Statement – I: The correct order in terms of atomic/ionic radii is

$Mg > Al > Mg^{+2} > Al^{+3}$

Statement – II: The correct order in terms of the magnitude of electron gain enthalpy is

$Cl > Br > S > O$

In the light of the above statement, choose the correct answer from the options given below

1) Both statement – I and statement – II are false

2) Both statement – I and statement – II are true

3) Statement – I is false but statement – II is true

4) Statement – I is true but statement – II is false

Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
Rankers Academy JEE
For More Join: @IITJEE_Advanced

54. **Assertion (A):** PH_3 is more basic than NH_3.

Reason (R): Electro negativity of nitrogen is more than phosphorus.

1) Both A and R are true and R is the correct explanation of A.

2) Both A and R are true and R is not the correct explanation of A.

3) A is true but R is false

4) A is false but R is true

55. The correct order of first ionization enthalpy values of the following elements in the order

A) O B) N C) Be D) F E) B

Choose the correct answer from the following options given below

1) $A > E > C > D > B$ **2)** $E > C > D > B > A$

3) $D > A > B > E > C$ **4)** $D > B > A > C > E$

56. $\wedge_{eq}$ of 0.01N solution of CaI_2 is $100.0\,S\,cm^2\,eq^{-1}$ at $298K$. Cell constant is $0.25\,cm^{-1}$, the current (in A) will flow when potential difference between electrode is 5V.

1) $0.10A$ **2)** $0.50\,A$ **3)** $0.02\,A$ **4)** $1A$

57. Consider the following first order completing reaction

$X \xrightarrow{K_1} A + B$ and $Y \xrightarrow{K_2} C + D$

If 50% of the reaction of 'X' was completed, when 96% of the reaction of Y was completed, the ratio of their rate constants $\left(\frac{K_2}{K_1}\right)$ is

1) 4.06 **2)** 0.215 **3)** 1.1 **4)** 4.65

58. The pH of 0.1M solution of the following salts decreases in the order

1) $NaCl > NH_4Cl > NaCN > HCl$ **2)** $NaCN > NaCl > NH_4Cl > HCl$

3) $NaCN > NH_4Cl > NaCl > HCl$ **4)** $HCl > NaCl > NaCN > NH_4Cl$

59. $NaC\ell O_3$ is used even in spacecrafts to produce O_2. The daily consumption of pure O_2 by a person is 246 lit at 1atm, 300K. The number of moles of $NaClO_3$ required to produce O_2 for daily consumption of a person at 1 atm, 300k

$NaClO_{3(s)} + Fe_{(s)} \longrightarrow O_{2(g)} + NaCl_{(s)} + FeO_{(s)}$

$R = 0.082 L.atm.mol^{-1} k^{-1}$

1) 40 moles **2)** 10 moles **3)** 20 moles **4)** 30 moles

Sri Chaitanya Educational Institutions
Infinity Learn


For More Join: @IITJEE_Advanced

60. 2 mole of an ideal gas at $27^{\circ}C$ expands isothermally and reversibly from a volume of 4 litre to 40 litre. The work done (in KJ) by the gas is

1) $w=-28.72\,KJ$ **2)** $w=-5.736\,KJ$ **3)** $w=-11.488\,KJ$ **4)** $w=-4.988\,KJ$

61. In which of the following molecule/ion all the bonds are not equal?

1) XeF_4 **2)** BF_4^- **3)** C_2H_4 **4)** SiF_4

62. In glycogen branched chain polymer of $\alpha-D-$glucose units in which branching occurs by the formation of glycosidic linkage between

1) C_1-C_4 of $\alpha-D-$glucose units **2)** C_2-C_4 of $\alpha-D-$glucose units

3) C_1-C_6 of $\alpha-D-$glucose units **4)** C_2-C_6 of $\alpha-D-$glucose units

63. Which of the following options represents the correct bond order

1) $O_2^+ > O_2^{-2} > N_2$ **2)** $N_2 > O_2^{-2} > O_2^+$ **3)** $N_2 > O_2^- > O_2^{-2}$ **4)** $O_2^+ > N_2 > O_2^{-2}$

64. Complex "X" of composition $Cr(H_2O)_6\,Cl_n$ has a spin only magnetic moment of 3.83 BM. It reacts with $AgNO_3$ and shows geometrical isomerism. The complex is

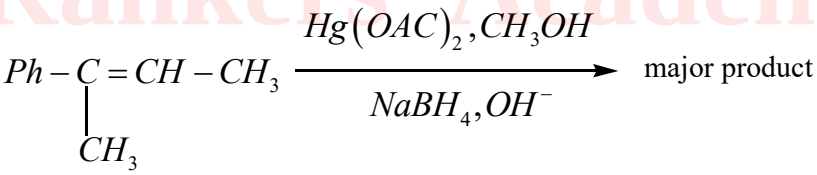
1) $\left[Cr(H_2O)_6\right]Cl_3$ **2)** $\left[Cr(H_2O)_4\,Cl_2\right]Cl_2.2H_2O$

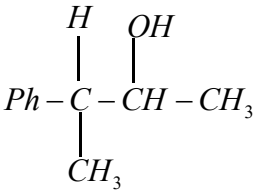
3) $\left[CrCl_2(H_2O)_4\right]Cl_2.2H_2O$ **4)** $\left[Cr(H_2O)_4\,Cl_2\right]Cl.2H_2O$

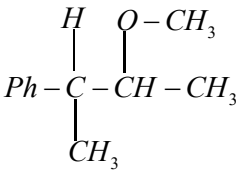
65. The CFSE for octahedral complex $\left[CoF_6\right]^{4-}$ is $18000\,cm^{-1}$. The CFSE for tetrahedral complex $\left[CoF_4\right]^{-2}$ will be

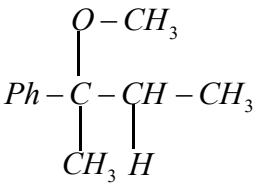
1) $16000\,cm^{-1}$ **2)** $8000\,cm^{-1}$ **3)** $18000\,cm^{-1}$ **4)** $20000\,cm^{-1}$

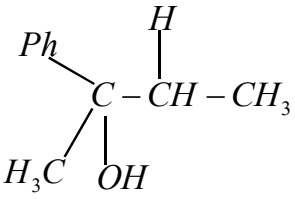
66. In the following reaction major product is



1) 

2) 

3) 

4) 

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

67. A first row transition metal (M) doesn't liberate H_2 gas from dilute HCl 1 mole of aqueous solution of MSO_4 is treated with excess of aqueous KCN and then $H_2S_{(g)}$ is passed through solution. The amount of MS (metal sulphide) formed from the above reaction is ___mole

1) 1 **2)** 0 **3)** 3 **4)** 2

68. The correct order of reactivity of CH_3Br in methanol with the following nucleophiles is $I^-, C_2H_5O^-$ & $C_6H_5O^-$.

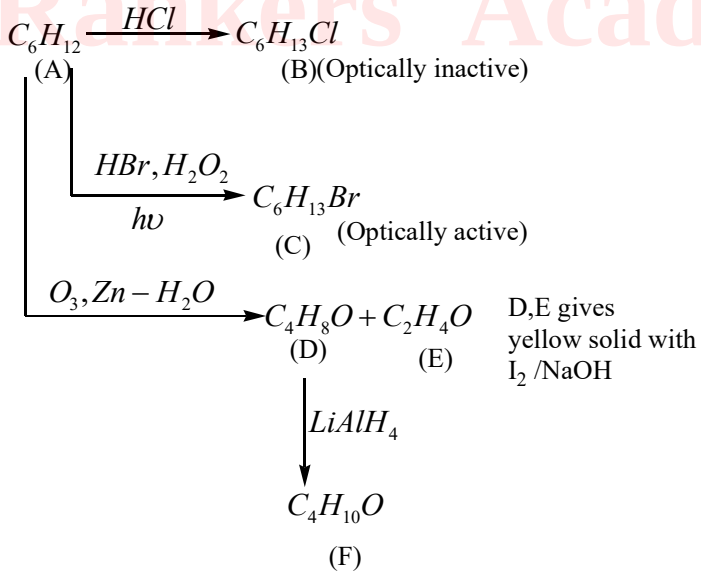
1) $I^- > C_2H_5O^- > C_6H_5O^-$ **2)** $C_2H_5O^- > I^- > C_6H_5O^-$

3) $C_6H_5O^- > C_2H_5O^- > I^-$ **4)** $I^- > C_2H_5O^- = C_6H_5O^-$

69. **Statement–I:** Aldehyde and terminal alkyne both react with Tollen's reagent.
Statement–II: Aldehyde and ketone can be distinguish by 2, 4– DNP.

1) Statement–I is correct and Statement–II is incorrect
2) Statement–I is incorrect and Statement–II is correct
3) Both Statement–I & Statement–II are correct
4) Both Statement–I & Statement–II are incorrect

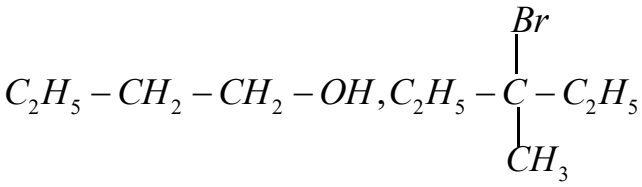
70. Identify C and F respectively in the following reaction.



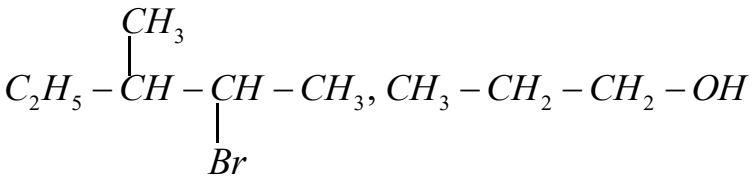
Rankers Academy JEE

Sri Chaitanya Educational Institutions
Infinity Learn
41

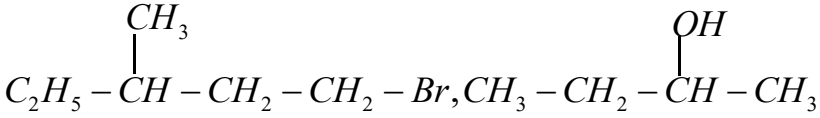

For More Join: @IITJEE_Advanced



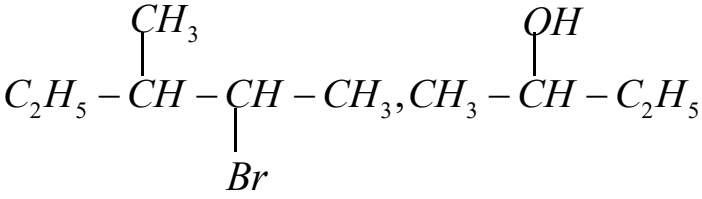
1)



2)



3)



4)

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

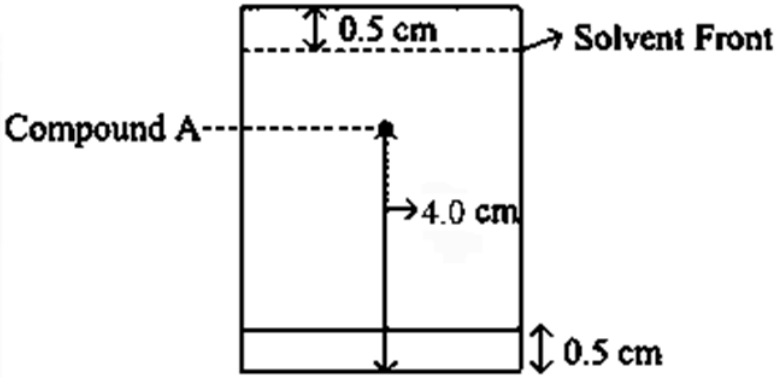
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. $\Delta H_{vap} = 30\,KJ/mol$ and $\Delta S_{vap} = 75\,J\,mol^{-1}K^{-1}$, then the temperature of vapour at one atmosphere in K is ___

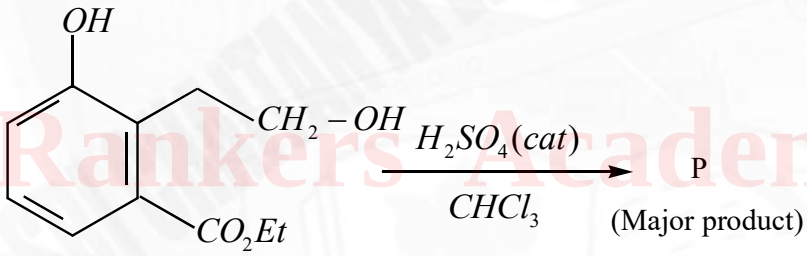
72. 2.0 g of polybasic organic acid (molar mass = 600) required 100 ml of a $\frac{M}{6}NaOH$ solution for complete neutralisation then basicity of acid is ____

Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

73. The following chromatogram was developed by adsorption of compound A on a 8cm TLC glass plate. Retardation factor of compound A is _____ $x \times 10^{-1}$ then the value of 'x' is



74. The number of oxygen atoms present in the "P" is x and number of sp^2 carbons are y, then the value of $\frac{x+y}{2}$ is



75. 0.25 gr of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of $CaCl_2$ tube and potash tube at the end of the experiment. The amount was found to be 0.15 gr, 0.1836 gr respectively.

The percentage of oxygen in compound A is ____% (Nearest integer)

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

Sri Chaitanya Educational Institutions | Infinity Learn

41 Years of Excellence 1986-2026

LEADING BY MILES SRI CHAITANYA DOMINATES

JEE ADVANCED 2025

29 Ranks in Top 100 in All-India Open Category



4 Students in Top 11 in JEE-Advanced 2025, All India Open Category

Rankers Academy JEE

16 RANK DEVDUTTA MAJHI HT. No. 255053116*	18 RANK DHARMANA GNANA RUTVIK SAI HT. No. 256055278	19 RANK VANGALA AJAY REDDY HT. No. 256131009	23 RANK AKSH GOGI HT. No. 252071075*	26 RANK P HEMA SAI SURYA KARTHIK HT. No. 256033006	27 RANK SARKARSAMUDRA HT. No. 252071105*
30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048245*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252061477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256166048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	745	NUMBER OF QUALIFIED RANKS	4,212

040 66 15 15 15 | www.srichaitanya.net

For More Join: @IITJEE_Advanced